

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Pipeline Integrity Monitoring/Management Philosophy.

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Revision History

Rev No.	Date	Description of Revision
R00	10 Aug 2022	Issued For Interdisciplinary Checks
R01	10 Aug 2022	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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Summary Page

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PPL-PHY-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

Abbreviations

Abbreviations	Meaning
ASME	American Society of Mechanical Engineers
API	American Petroleum Institute
CS	Carbon Steel
DEP	SPDC- Design and Engineering Practices
DP	Design Pressure
DWT	Drop Weight tear
FEED	Front End Engineering
HAZ	Heat affected zone
ISO	International Organisation for Standardization
LTO	License To Operate
3LPE	Three-Layer Polyethylene
MMscf/d	Million standard cubic feet per day
NDE	Non-Destructive Examination
NPS	Nominal Pipe Size
OD	Outer Diameter
PE	Polyethylene
PSL	Product Specification Level
TD	Design Temperature
SPDC	Shell Petroleum Development COMPANY Nig. Limited
SMTS	Specified Minimum Tensile Strength
SMYS	Specified Minimum Yield Strength
WT	Wall thickness
WPS	Welding Procedure Specification
WPQR	Welding Procedure Qualification Record

Definition of Terms

Terms	Definitions
BFPCL	Brass Fertilizer & Petrochemical Limited.
COMPANY	Party (Usually BFPCL) that initiates the project and ultimately pays for its design and construction. The COMPANY will generally specify the technical requirements. The COMPANY may also include an agent or consultant authorized to act for and on behalf of, the COMPANY.
Engineer	The party that carries out the FEED Engineering of the project.
Manufacturer/Supplier	Party, which manufactures or supplies equipment and services to perform the duties specified by the MANUFACTURER.
Shall	The word 'shall' indicate a mandatory requirement.
Should	The word 'should' indicate a recommendation.
Project	Front End Engineering Design for Gas Gathering Pipelines (Phase1)

Site and Environmental Conditions

Refer to the document EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002_Basis of Design

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1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

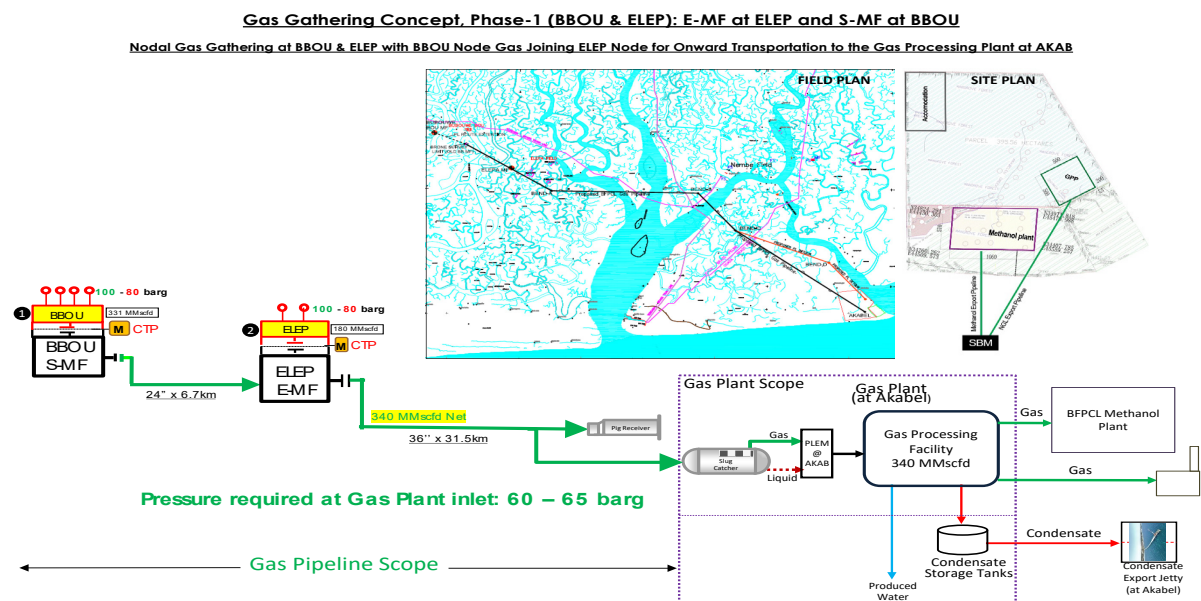


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

As part of the operation and maintenance philosophy, this document defines and specifies the minimum requirements for the integrity monitoring and management of BFPCL Gas Gathering Pipelines (Phase1) system.

This Philosophy provides the basis for defining the minimum inspection requirements necessary to assure the integrity of Pipeline installations and structures with the scope of this project.

It provides the optimum level of inspection to assure the integrity of the installations and therefore these minimum requirements should only be seen as a safety net below which integrity standards must not fall.

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendment or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- COMPANY's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the COMPANY for resolution.

2.1. Project Technical Documents

Key Project Technical Specifications are listed below.

Ref	Document Number	Title
1	EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
2	EA-BFPCL-GPMC-GGPL-FEED-GEN-PHY-001	Pipeline Design & Operating Philosophy
3	EA-BFPCL-GPMC-GGPL-FEED-OPS-PHY-001	Operations and Maintenance Philosophy
4	EA-BFPCL-GPMC-GGPL-FEED-PSE-PHY-002	Process Control & Safeguarding Philosophy
5	EA-BFPCL-GPMC-GGPL-FEED-MCM-DOC-001	Material Specification and Corrosion Management
6	EA-BFPCL-GPMC-GGPL-FEED-PSE-PLN-003	Pipeline Pre-commissioning Specification

2.2. Shell Specifications

The following DEPs are applicable:

Ref	Document Number	Title
1	DEP. 82.00.10.10-Gen	Project Quality Assurance
2	DEP. 31.40.40.38-Gen	Hydrostatic Pressure Testing of new Pipeline (February 2017)
3	DEP. 61.40.20.30-Gen	Welding of Pipelines & Related Facilities (Amendments / supplements to API Std 1104)
4	DEP 31.40.40.38-Gen	Hydrostatic pressure testing of new pipelines
5	DEP 31.40.60.11-Gen.	Pipeline leak detection
6	DEP 31.40.50.30-Gen	Pre-commissioning of pipelines

2.3. Local Regulations

S/N	Regulation
1	Petroleum Act, CAP P10, Laws of the Federation of Nigeria, LFN 2004, amended.
2	Mineral Oil (Safety) Regulations 1963, as amended.
3	Oil Pipelines Act, 1956. CAP. 145 (Amended 1965, CAP. 338, LFN 1990)
4	Oil and Gas Pipelines Regulations, 1995
5	Environmental Impact Assessment Act, CAP E12, LFN 2004.
6	Federal Environment Protection Agency Act, CAP F13, LFN 2004.
7	Environmental Guidelines and Standards for The Petroleum Industry In Nigeria, 1991.

2.4. International Codes and Standards

Ref	Document Number	Title
1	ASME B31.8	Gas Transmission and Distribution Piping Systems
2	ISO 9001	Quality Systems Model for Quality Assurance in Design, Development, Production, Installation and Servicing
3	API RP 1111	- Design, Construction, Operation & Maintenance of Offshore Hydrocarbon Pipelines - 1999

3. Line Pipe Data

3.1. 24" Multiphase Pipeline Data

The 24" pipeline design data for the Multiphase Pipeline Project are as shown below

Parameter	Description
Design Code	ASME B31.8
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	6.7km
Design Life	30 years
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Line Pipe Material Grade	API 5L-X70(PLS2) SAWL - 100% radiographed.
NPS	24"
Wall Thickness	15.88mm
Design Factor	0.6
Corrosion Allowance	3 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	60mm
Installation Condition	Trenching
Cathodic Protection	Impressed current

3.2. 36" Multiphase Pipeline Design Data

The 36" pipeline design data for the Multiphase Pipeline Project are as shown below

Parameter	Description
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Design Life	30 years
Process Fluid	Condensate/hydrated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Line Pipe Material Grade	API 5L-X70 (PLS2) SAWL - 100% radiographed.
NPS	36"
Wall Thickness	23.83mm
Design Factor	0.6
Corrosion Allowance	6 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	85mm
Installation Condition	Trenching
Cathodic Protection	Impressed current

4. Risk Based Inspection

In order to focus inspection effort in the most efficient manner, a Risk Based Inspection (RBI) approach is the integrity management strategy recommended for the major equipment and structures of the pipeline system. Where this approach is approved and adopted, its output replaces the minimum requirements specified in this philosophy.

However, where an approved formal RBI approach is NOT applied then the minimum inspection requirements identified herein should be adhered to.

Attention is raised that an inspection plan designed according to RBI may be more stringent for some equipment and less for others depending on their criticality, than the requirements set out in this document.

5. Integrity Assessment.

5.1. Objectives

The goal of the pipeline integrity monitoring and management is to preventing structural integrity problems that may develop on the pipeline from having a significant effect on the host communities, the environment and the business objectives by identifying and carrying out effective inspection, monitoring and repairs on the pipeline system as at when due.

A responsive and well-planned pipeline integrity management programme shall be developed in the detailed engineering phase taken into account all in the identified risk inherent and associated with the pipeline system including the recommendations pointed out in the environmental assessment studies carried on the pipeline project.

5.2. Pipeline Failures.

The primary causes of pipeline failures in Nigeria which anecdotally apply this pipeline system are as follows;

- Encroachment and contact damage by third party (Oil Theft, Population encroachment etc).
- Internal and external Corrosion - Pitting and scaling of the surface of the line pipe.
- Geotechnical failures due to the soil characteristics of the ROW;
 - Soil Movement,
 - Land Slide,
 - Floods etc.
- Stress Corrosion agents such as Corbon dioxide (CO₂) and sulphides.H₂S
- Pipeline material, equipment failure and operators' errors.

5.3. Methodology.

The following method shall be considered in the assessment of the integrity of the pipeline system.

- Regular Visual inspection of the Right of Way (ROW) to identify potential area of encroachment and ROW degradations or failures.
- Depth of the Pipeline survey by using scanners or manual sport excavation to identify possible arears of settlements.
- External non-destructive testing (NDT)
 - Radiography
 - Magnetic particular testing
 - Dye penetrant Testing
 - Ultrasonic inspection
- Cathodic protection monitoring

- Coating disbondment and damage survey
- Hydrostatic Testing
- Geometry in-line inspection (GILI) tools such as intelligent and calliper pigs, X,Y,Z geometric (inertial Guidance) tools
- Ultrasonic in-line inspection tools such as the conventional

6. Integrity Assessment & Monitoring Requirements

6.1. General

The minimum acceptable inspection requirements to assure the integrity of the Mainlines (24" and 36") Pipeline systems are given in Table 6.0. These requirements have been prepared general for the 24" BBOU to ELAPA pipeline System and 36" ELAPA to AKABEL pipeline system. This document should be further developed in detailed engineering when additional survey data on the ROW is available.

Specific written schemes of examination shall be prepared for each individual pipeline from pig launcher to receiver even although the pig traps themselves can be assessed as Pressure Vessels even as they are designed as per ASME B 31.8. Since both the consequence and probability of failure will change over the length of each pipeline, the first task in preparing inspection programmes shall be to sub-divide each line into distinct portions. Demarcation of each portion shall be drawn to reflect changes in consequence and probability of failure.

Risk Based Inspection (RBI) approach is considered to be best option for the pipelines. Where this approach is considered, a criticality assessment should be carried out to identify the consequence and probability of failure for each pipeline portion. Therefore, its output and findings shall replace the minimum requirements specified herein in Table 6.0. Conversely the RBI schemes may increase the inspection requirements especially as pipelines is carrying the gas which are deemed critical in terms of economic impact on the Brass fertilizer project.

The output from criticality assessments or the minimum requirements specified in Table 6.0, together with local legislation listed above shall then enable the formulation of a written scheme of examination for each Pipeline system.

6.2. Design Review

The Pipeline systems shall be designed to allow their integrity to be verified and monitored throughout their 30years operational life span. Generally, this will mean they shall be designed to allow both **Operational and Intelligent Pigging**.

Pipeline bend radii of **5xDiameter** of the pipeline shall be sufficient to allow passage of intelligent pigs, only barred tees shall be installed on pipelines and pig traps. Sufficiently long horizontal pig traps shall be installed with adequate lifting facilities and access space to allow the handling and insertion of intelligent pigs. Lay-down areas or deck hatches shall also be installed to facilitate pig handling. Sufficient pig detection equipment shall also be installed at BBOU, ELEPA platforms and AKABEL Piping station.

Pipeline markers capable of detection by intelligent pig should be installed at regular intervals along the pipeline. These may simply be short pup pieces.

6.3. Emergency Pipeline Repair Schemes (EPRS)

Future Emergency Pipeline Repair Schemes (EPRS) shall be adopted as part of the pipeline integrity management system. Adequate surplus materials must be procured at the project stage to ensure that an emergency repair can be performed. Adequate resources and Personnel shall also be kept on ground to execute any emergency repairs on the Pipelines.

6.4. Corrosion Monitoring

Due to the presence of CO₂ in the composition of the effluent to be transported by the pipeline systems, there is a great risk of internal corrosion on the entire pipeline system.

A continuous corrosion inhibitor injection system (chemical injection) is proposed to mitigate the CO₂ corrosion though, there is need to install a combination of corrosion probes to monitor the effectiveness of this mitigation and the rate of corrosion of the pipeline

Chemical analysis shall be utilised to give a general indication of the corrosive behaviour of the treated effluent to continuously adjust the requirement of the chemical injection if need be. The information gleaned from corrosion monitoring techniques shall be used together with inspection results to help determine preventative treatment and the efficiency of such treatment.

The type of corrosion anticipated shall determine the number, type and position of probes installed together with any other chemical analysis. Specific programmes shall be prepared to determine the frequency at which monitored shall be performed.

Different options are available in the form of integrated monitoring system where real-time life transmission of the corrosion rate is acquired or the manual retractable probes and coupons shown below that are retracted from the pipeline on routine bases for analysis. The choice of the corrosion monitoring system shall be finalised in the detailed engineering phase.



Manual retractable corrosion probes

Corrosion Monitoring readings and subsequent analysis shall be performed by either the Integrity Department or the Laboratory/Treatment Department of Company.

Corrosion monitoring devices shall be installed at pipeline ends at a position representative of the anticipated corrosion. On pipelines susceptible to Top of Line (TOL) Corrosion consideration shall be given to installing corrosion monitoring devices such as Ultrasonic Mats on pipeline doglegs. The interrogation and monitoring devices shall be located on the adjacent platform. The primary function of such devices shall be to detect TOL corrosion and assess the effectiveness of chemical treatment.

6.5. Inspection Scope

A range of differing inspections shall be performed on Pipelines but intelligent pigging is the primary means for assuring pipeline integrity.

6.5.1. *Pigging*

Due to a number of recent case histories within Nigeria, it is advised that these pipelines should be intelligently pigged prior to start-up in order to provide a baseline reference for future inspections. In addition to intelligent pigging, gauging and calliper pigs shall be run to prove the future pigability of the pipeline system. This will normally be the responsibility of the EPC Contractor and included in the pipeline commissioning scope of work.

The first in-service intelligent pigging run should be performed within two to three years of pipeline start-up depending on corrosion studies. Subsequent intelligent pig runs should be performed at a frequency not **exceeding 5 years**. Where detailed corrosion studies have been performed and the risk of internal corrosion mitigated to the barest minimum level, the frequency of intelligent pigging may be increased to a **maximum of 10 years** as far as regular external CP monitoring is performed and is effective.

6.5.2. *NDT*

The topsides or above ground sections of the pipelines shall be inspected to give a general inferred internal condition of the remaining pipeline. Both ultrasonic wall thickness measurements and corrosion monitoring techniques shall be utilised. These ultrasonic wall thickness measurements shall be performed at a maximum **frequency of 2 yearly**.

The frequency of corrosion monitoring shall be determined by a specific programme and could include monitoring by Electric Resistance probes, Corrosion Coupons, Iron Content and residual corrosion inhibitor. It should however be noted that this type of inspection will not identify all types of corrosion on the remaining pipeline and must be used in addition to intelligent pigging and as an alternative.

6.5.3. *Visual Inspection*

The riser portion of the pipeline as the approach the manifold platforms at BBOU and ELEPA and the underground-aboveground interface of the pipelines at AKABEL pigging station should be considered as the most critical areas for assessment. These zones shall be visually inspected at a maximum frequency of **2 yearly**. The use of abseiling techniques can be considered to prevent the need for expensive scaffold if these part of the is too high to reach from the platforms.

The Pipeline routes or Rights of Way (ROW) shall be routinely patrolled at a maximum frequency of **yearly** to identify any potential areas of concern. Where ROW is open to the public such as road crossings, patrols shall be performed at a maximum frequency of **6 monthly**.

Pipeline river crossings shall be surveyed at a maximum frequency of **2 yearly** to assure adequate burial by either acoustic or magnetic means. For pipelines containing inert fluids, river crossings shall be inspected at a maximum frequency of **5 years**.

6.5.4. *Cathodic Protection*

Cathodic Protection (CP) systems shall be designed with test points along the entire pipeline system lines. The CP Systems shall allow on/off potential surveys to be performed as well.

Cathodic Protection (CP) shall be monitored over the complete length of all onshore pipelines at a maximum frequency of **2 yearly**.

The Pipeline Systems shall be protected by temporary sacrificial anode system which shall be decommissioned and replaced by a permanent impressed current CP system soon after commissioning of the impressed current CP system. The Impressed current CP systems shall be monitored using "instant - off" methods. The wastage of sacrificial anodes shall be assessed as part of any survey to predict when retro fits may be needed.

The output of the Transformer/Rectifiers (T/R) of the impressed current CP monitoring system should be monitored frequently **a monthly basis** because the performance of the transformer and the entire CP system is critical to the availability of the pipeline system to attain its designed life span. The performance of the CP systems should include the analysis of the T/R output.

6.5.5. *ESDs*

The performance of pipeline Emergency Shutdown Valves (ESDs) including HIPPS (if fitted) shall be monitored by means of functional testing at a maximum frequency of yearly. The tightness of such valves shall also be assessed. A performance standard shall be prepared and used to determine the acceptance criteria for the function & leak test. The execution of such tests shall normally be performed by the Production Department but are included in this referential for completeness.

Table 6.0: Pipeline Inspection Integrity Requirement

Inspection Type/Integrity Tasks	When	Rule (R) or Guidance (G)	Comments
Pre Start-Up Phase			
Construction Inspection			
Manufacture and hence QA of pipelines shall be the responsibility of EPC Contractor. This section is therefore limited to modification and repairs of pipelines. Individual Quality Plans shall be prepared for each modification or repair. The level of Quality Control shall be determined in accordance with the General Specification for Pipelines.			
Commissioning/Baseline Inspection			
Verification of Technical Records and Registration of Technical Details in Pipe-	Before Start-up	R	
Preparation of Inspection Drawings with component identification		R	
Verification survey of installed pipeline and as-built survey, river crossings and Right of Way.	During Survey	R	
General Visual Inspection of the risers shall be performed during as-installed survey of the platforms	During platform survey	R	
External General Visual Inspection of topsides and above ground sections and Wall Thickness Measurements (where internal corrosion is identified as a possible mode of failure)	Before Start-up	R	
Baseline Inspection by Intelligent Pig		G	
Review the baseline gauging and calliper survey of the pipeline, assess pigging facilities for future pigging operations and confirm tightness of trap isolation valves		R	
Witness Cathodic Protection baseline survey together with commissioning of impressed current transformer rectifi-		R	
OPERATIONAL PHASE			
First In-Service Inspection			
Land Pipeline Right of Way (ROW) Survey	Within 6Months or1 Year of start-	R	Public ROW - 6 Months COMPANY ROW -1Year

Inspection Type/Integrity Tasks	When	Rule (R) or Guidance (G)	Comments
External General Visual Inspection of Riser Zone and Atmospheric Section	Not > 2 yearly	R	
Wall Thickness Measurements (where internal corrosion is identified as a possible mode of failure) of topsides, riser and above ground sections	Not > 2 year from start-up	R	
Initial in-service intelligent pigging survey	Within 2 - 3 Years of start-up	R	To be determined by corrosion study.
500 metre zones, pipeline crossings and other pipeline features	Not > 2 year from start-up	R	
River Crossings - Magnetic, acoustic burial survey plus visual inspection at		R	
Cathodic Protection (CP) survey of complete pipeline		R	
ESD Valve performance testing	Not > 1 yearly	R	
SUBSEQUENT IN-SERVICE INSPECTIONS			
Land Pipeline Right of Way (ROW) Survey	6 Monthly or 1Yearly	R	Public ROW - 3 Monthly COMPANY ROW - Yearly
External General Visual Inspection of Splash Zone of platform structures	Not > 2 yearly		
External General Visual Inspection and Wall Thickness Measurements (where internal corrosion is identified as a possible mode of failure) of above-ground sections	Not > 2 yearly	R	
External General Visual Inspection of pipeline risers shall be performed during first in-service survey of the BBOU and ELEPA manifold Platforms	Not > 2 yearly	R	
Intelligent pigging survey	Not > 5 yearly	R	
River Crossings - Magnetic, acoustic burial survey plus visual inspection at low tide/dry season	Not > 2 yearly	R	
Cathodic Protection (CP) survey of complete onshore pipeline	Not > 2 yearly	R	
Cathodic Protection (CP) survey of complete pipeline	Not > 5 yearly	R	May be extended to 10 yearly if the records of the previous survey did not show any significant deteriorations.

Inspection Type/Integrity Tasks	When	Rule (R) or Guidance (G)	Comments
ESD Valve performance testing	Not > 1 yearly	R	
Inspection Reviews			
Following a significant change in the environment conditions, extreme environmental events, dropped objects, vessel collisions, observed damage		G	